Started on	Friday, 9 October 2026, 10:37 AM
State	Finished
Completed on	Friday, 9 October 2026, 10:45 AM
Time taken	8 mins 21 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
int main()
{
int n;
scanf("%d",&n);
int arr[n];
for(int i=0;i<n;i++)
{
scanf("%d",&arr[i]);

}
int dp[n];
for(int i=0;i<n;i++)
{
    dp[i]=1;
}

for(int i=1;i<n;i++)
{
for(int j=0;j<i;j++)
{
if(arr[j]<=arr[i]){
if(dp[j]+1>dp[i])
dp[i]=dp[j]+1;
}
}
}
int ans=0;
for(int i=0;i<n;i++)
{
if(dp[i]>ans)
ans=dp[i];
}
printf("%d",ans);
}
```

	Input	Expected	Got	
✔	9 -1 3 4 5 2 2 2 2 3	6	6	✔
✔	7 1 2 2 4 5 7 6	6	6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.